

MAE 277 Course Project - Developing a Model and Applying Infinite-Horizon, Continuous LQR

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1 Developing the Model

The project will be focusing on non-isothermal CSTRs in series, the first one containing the second-order, exothermic reaction R_1 , $A + B \rightarrow C$; the second one containing R_1 and the second-order, exothermic reaction R_2 , $C + D \rightarrow E$. The inlet flow rates (F_{A0} , F_{B0} , F_{D0} , F_x , and F_y) are the manipulated variables—relating this to a real life system, consider that we are manipulating the valves outlined in blue.

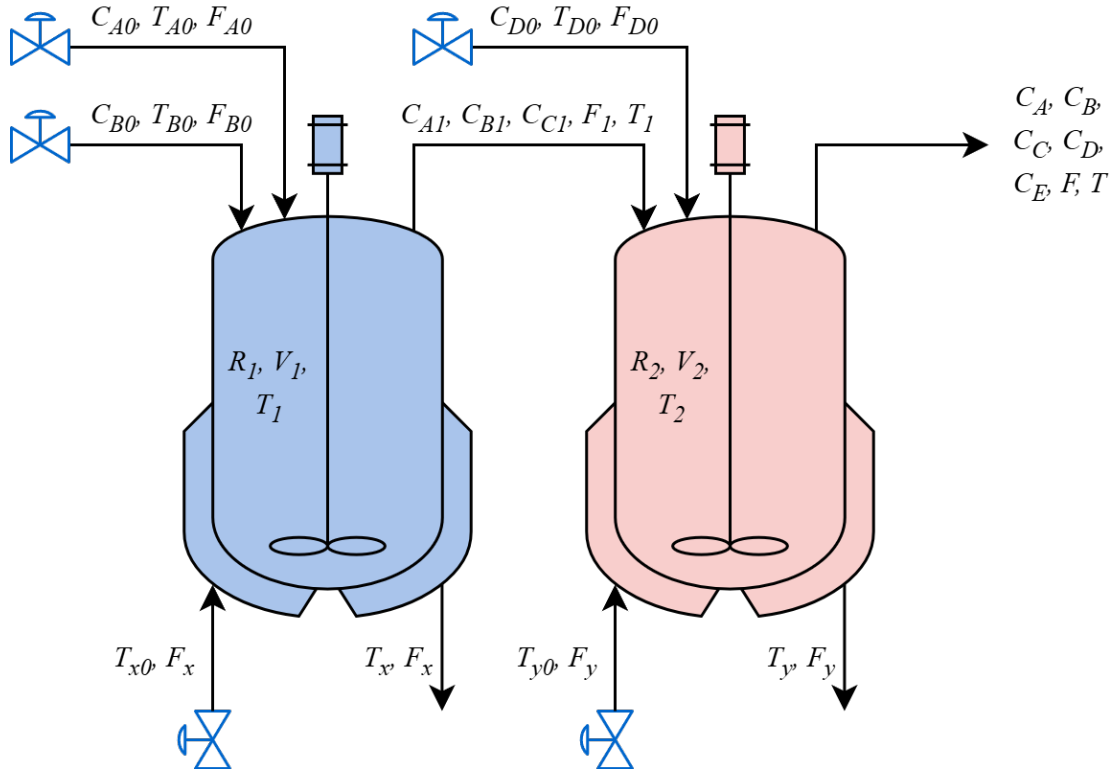


Figure 1: Two non-isothermal CSTRs in series with stream variables listed.

For each of the two reactors, a mole balance for each species and an energy balance will need to be performed to accurately model the nonlinear behavior. For reference, the general mass balance equation

(the so-called "most important equation for chemical engineers") is summarized as,

$$\text{Accumulation} = \text{In} - \text{Out} + \text{Generation} - \text{Consumption},$$

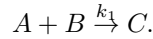
and serves as the basis for the model. Unless otherwise stated, the following assumptions will be made:

- The chemical reaction rate law follows the power law (the n -th reaction has the reaction rate $r_n = k_n \prod_i C_i^{\nu_i}$)
- The reaction rate constant k follows the Arrhenius equation $k_n = A_n \exp(E_n/RT)$
- The chemical reaction is strictly aqueous and no solidification/crystallization/evaporation occurs
- Both CSTRs are well-mixed (no spatial dependence for concentration or reactor temperature)
- The cooling jacket is well mixed (no spatial dependence for jacket temperature)
- All reactive species are dilute and make negligible impact on the heat capacity of the fluid mixture
- Heat capacity is constant with temperature

With this, the dynamical model is (relatively) straightforward to develop. Part of the model formulation took inspiration from Shyamalogowri and Rajeswari's paper on CSTR modeling [1]

1.1 CSTR #1

Recall that the chemical reaction in CSTR #1 is R_1 ,



The model for the first CSTR is below. All variables, their descriptions, and possible values to use in simulations will be listed at the end of the document.

$$\frac{dC_{A1}}{dt} = \left(\frac{F_{A0}C_{A0}}{V_1} \right) - \left(\frac{F_1}{V_1} \right) C_{A1} - A_1 e^{-\frac{E_1}{RT_1}} C_{A1}C_{B1} \quad (1)$$

$$\frac{dC_{B1}}{dt} = \left(\frac{F_{B0}C_{B0}}{V_1} \right) - \left(\frac{F_1}{V_1} \right) C_{B1} - A_1 e^{-\frac{E_1}{RT_1}} C_{A1}C_{B1} \quad (2)$$

$$\frac{dC_{C1}}{dt} = - \left(\frac{F_1}{V_1} \right) C_{C1} + A_1 e^{-\frac{E_1}{RT_1}} C_{A1}C_{B1} \quad (3)$$

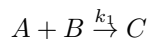
$$\frac{dT_1}{dt} = \left(\frac{F_{A0}T_{A0}}{V_1} \right) + \left(\frac{F_{B0}T_{B0}}{V_1} \right) - \left(\frac{F_1}{V_1} \right) T_1 + \frac{(-\Delta H_1)}{\rho_r C_{pr}} A_1 e^{-\frac{E_1}{RT_1}} C_{A1}C_{B1} - \frac{U_x A_x}{\rho_r C_{pr} V_x} (T_1 - T_x) \quad (4)$$

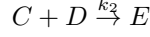
$$\frac{dT_x}{dt} = \frac{U_x A_x}{\rho_c C_{pc} V_x} (T_1 - T_x) + \frac{F_x}{V_x} (T_{x0} - T_x) \quad (5)$$

where $F_1 = F_{A0} + F_{B0}$.

1.2 CSTR #2

Recall that the chemical reactions in CSTR #2 are R_1 and R_2 ,





The model for the second CSTR is below. Again, all variables will be listed at the end of the document.

$$\frac{dC_A}{dt} = \left(\frac{F_1 C_{A1}}{V_2} \right) - \left(\frac{F}{V_2} \right) C_A - A_1 e^{-\frac{E_1}{RT}} C_A C_B \quad (6)$$

$$\frac{dC_B}{dt} = \left(\frac{F_1 C_{B1}}{V_2} \right) - \left(\frac{F}{V_2} \right) C_B - A_1 e^{-\frac{E_1}{RT}} C_A C_B \quad (7)$$

$$\frac{dC_C}{dt} = \left(\frac{F_1 C_{C1}}{V_2} \right) - \left(\frac{F}{V_2} \right) C_C - A_2 e^{-\frac{E_2}{RT}} C_C C_D + A_1 e^{-\frac{E_1}{RT}} C_A C_B \quad (8)$$

$$\frac{dC_D}{dt} = \left(\frac{F_1 C_{D0}}{V_2} \right) - \left(\frac{F}{V_2} \right) C_D - A_2 e^{-\frac{E_2}{RT}} C_C C_D \quad (9)$$

$$\frac{dC_E}{dt} = - \left(\frac{F}{V_2} \right) C_E + A_2 e^{-\frac{E_2}{RT}} C_C C_D \quad (10)$$

$$\frac{dT}{dt} = \left(\frac{F_1 T_1}{V_2} \right) + \left(\frac{F_{D0} T_{D0}}{V_2} \right) - \left(\frac{F}{V_2} \right) T + \frac{(-\Delta H_1)}{\rho_r C_{pr}} A_1 e^{-\frac{E_1}{RT}} C_A C_B + \frac{(-\Delta H_2)}{\rho_r C_{pr}} A_2 e^{-\frac{E_2}{RT}} C_C C_D \quad (11)$$

$$- \frac{U_y A_y}{\rho_r C_{pr} V_y} (T - T_y) \quad (12)$$

$$\frac{dT_y}{dt} = \frac{U_y A_y}{\rho_c C_{pc} V_y} (T - T_y) + \frac{F_y}{V_y} (T_{y0} - T_y) \quad (13)$$

where $F = F_{A0} + F_{B0} + F_{D0}$. We can plot the solutions to these ODEs using the values listed at the end of the document to verify that they appear at least somewhat accurate. Figure (2) shows that the behavior of the solutions to these ODEs do indeed match what our intuition would suggest.

1.3 State and Control Vectors

Since we need the concentrations and temperatures from both reactors, we need all of their states in our state vector. We are also manipulating the inlet volumetric flow rates for the species A , B , D , x , and y (x and y are the cooling fluid for the jackets for CSTR #1 and #2, respectively). Thus the state and control vectors are,

$$\mathbf{x} = \begin{bmatrix} C_{A1} & C_{B1} & C_{C1} & C_A & C_B & C_C & C_D & C_E & T_1 & T & T_x & T_y \end{bmatrix}^\top,$$

$$\mathbf{u} = \begin{bmatrix} F_{A0} & F_{B0} & F_{D0} & F_x & F_y \end{bmatrix}^\top.$$

2 Infinite-Horizon, Continuous LQR

2.1 Steady-State Values of State Variables

Since LQR requires that the dynamical model is linear, we must begin the linearization process. First we will need to determine the steady-state values of all state variables around a set of steady-state input values. We can begin by setting all time derivatives equal to zero, then solve for the steady-state values. This task can also be done entirely numerically, where the steady-state values are solved for using a root-finding function and setting the state evolution functions equal to zero. Doing this we obtain the $\mathbf{x}_{eq}$ vector, where

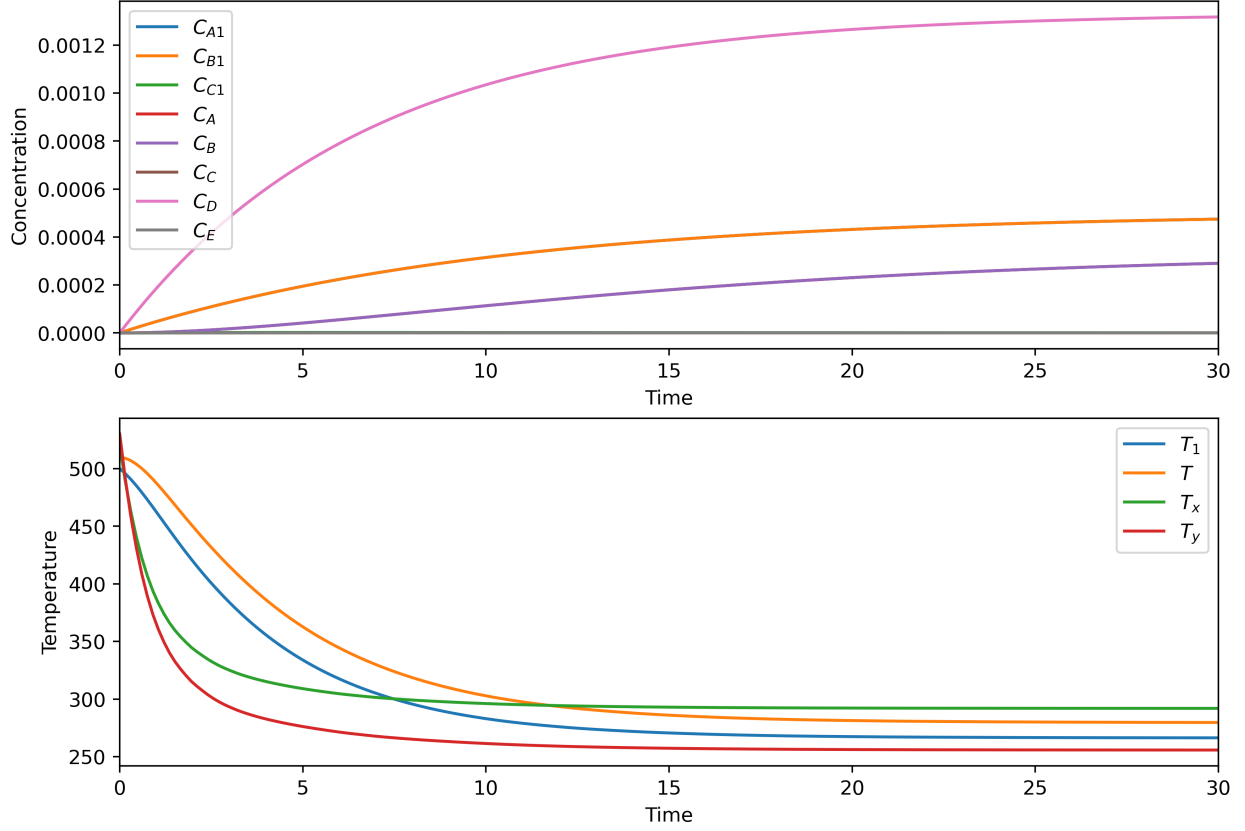


Figure 2: Concentration and temperature profiles for each species and vessel.

concentrations have units of mol/m^3 and temperatures have units of Kelvin.

2.2 Linearization of Model

We next need to linearize the model to put it into state-space form. We define the deviation state and control vectors,

$$\mathbf{x}' = \mathbf{x} - \mathbf{x}_{eq}, \quad \mathbf{u}' = \mathbf{u} - \mathbf{u}_{eq},$$

however for brevity, we will use the (somewhat abusive) notation of $\mathbf{x}' = \mathbf{x}$ and $\mathbf{u}' = \mathbf{u}$, and this notation will hold true for the remainder of the document. With this, we aim to derive the state-space model and linear control law of the form,

$$\dot{\mathbf{x}}(t) = A\mathbf{x}(t) + B\mathbf{u}(t), \tag{14}$$

$$\mathbf{u}(t) = -K\mathbf{x}(t). \tag{15}$$

The state matrix A and input matrix B are obtained by finding the Jacobian matrices of each state evolution equation $f_n(\mathbf{x})$ with respect to the states $\mathbf{x} \in \mathbb{R}^n$ and inputs $\mathbf{u} \in \mathbb{R}^m$, respectively,

$$A = \frac{\partial f}{\partial \mathbf{x}} \Big|_{\mathbf{x}_{eq}, \mathbf{u}_{eq}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} \Big|_{\mathbf{x}_{eq}, \mathbf{u}_{eq}} & \frac{\partial f_1}{\partial x_2} \Big|_{\mathbf{x}_{eq}, \mathbf{u}_{eq}} & \dots & \frac{\partial f_1}{\partial x_n} \Big|_{\mathbf{x}_{eq}, \mathbf{u}_{eq}} \\ \frac{\partial f_2}{\partial x_1} \Big|_{\mathbf{x}_{eq}, \mathbf{u}_{eq}} & \frac{\partial f_2}{\partial x_2} \Big|_{\mathbf{x}_{eq}, \mathbf{u}_{eq}} & & \vdots \\ \vdots & & \ddots & \\ \frac{\partial f_n}{\partial x_1} \Big|_{\mathbf{x}_{eq}, \mathbf{u}_{eq}} & \dots & & \frac{\partial f_n}{\partial x_n} \Big|_{\mathbf{x}_{eq}, \mathbf{u}_{eq}} \end{bmatrix}, \quad (16)$$

$$B = \frac{\partial f}{\partial \mathbf{u}} \Big|_{\mathbf{x}_{eq}, \mathbf{u}_{eq}} = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} \Big|_{\mathbf{x}_{eq}, \mathbf{u}_{eq}} & \frac{\partial f_1}{\partial u_2} \Big|_{\mathbf{x}_{eq}, \mathbf{u}_{eq}} & \dots & \frac{\partial f_1}{\partial u_m} \Big|_{\mathbf{x}_{eq}, \mathbf{u}_{eq}} \\ \frac{\partial f_2}{\partial u_1} \Big|_{\mathbf{x}_{eq}, \mathbf{u}_{eq}} & \frac{\partial f_2}{\partial u_2} \Big|_{\mathbf{x}_{eq}, \mathbf{u}_{eq}} & & \vdots \\ \vdots & & \ddots & \\ \frac{\partial f_n}{\partial u_1} \Big|_{\mathbf{x}_{eq}, \mathbf{u}_{eq}} & \dots & & \frac{\partial f_n}{\partial u_m} \Big|_{\mathbf{x}_{eq}, \mathbf{u}_{eq}} \end{bmatrix}. \quad (17)$$

Since these matrices are going to be quite large, the Jacobian can be numerically approximated using the central difference formula,

$$\frac{\partial f}{\partial x} \approx \frac{f(x + \Delta x) - f(x - \Delta x)}{2\Delta x}. \quad (18)$$

2.3 Calculate Optimal Gain

The gain for infinite-horizon, continuous LQR, K , is one such that the cost function

$$J = \int_0^\infty (\mathbf{x}^\top Q \mathbf{x} + \mathbf{u}^\top R \mathbf{u}) dt \quad (19)$$

is minimized over the control input $\mathbf{u}$. The state cost matrix Q and control cost matrix R are chosen by the control designer. The value of K can be calculated through the equation,

$$K = R^{-1} B^\top P, \quad (20)$$

where P is the solution to the continuous-time algebraic Riccati equation (CARE),

$$A^\top P + PA - PBR^{-1}B^\top P + Q = 0. \quad (21)$$

3 Results and Discussion

The simulation was completed in Python using Numpy for array definition and Scipy for linear algebra computations, Jacobian calculation, and ODE integration. The results of this simulation are in figure (3). From the figure we can see that the LQR controller is indeed driving the states towards their equilibrium values (dashed lines in the concentration and temperature plots).

An additional interesting detail we can see in the plots is that the concentrations dip below zero, and several concentration/temperatures profiles see dramatic rates of change in their values. This is because LQR is an unconstrained control method, and whatever it determines is optimal is what gets applied—in

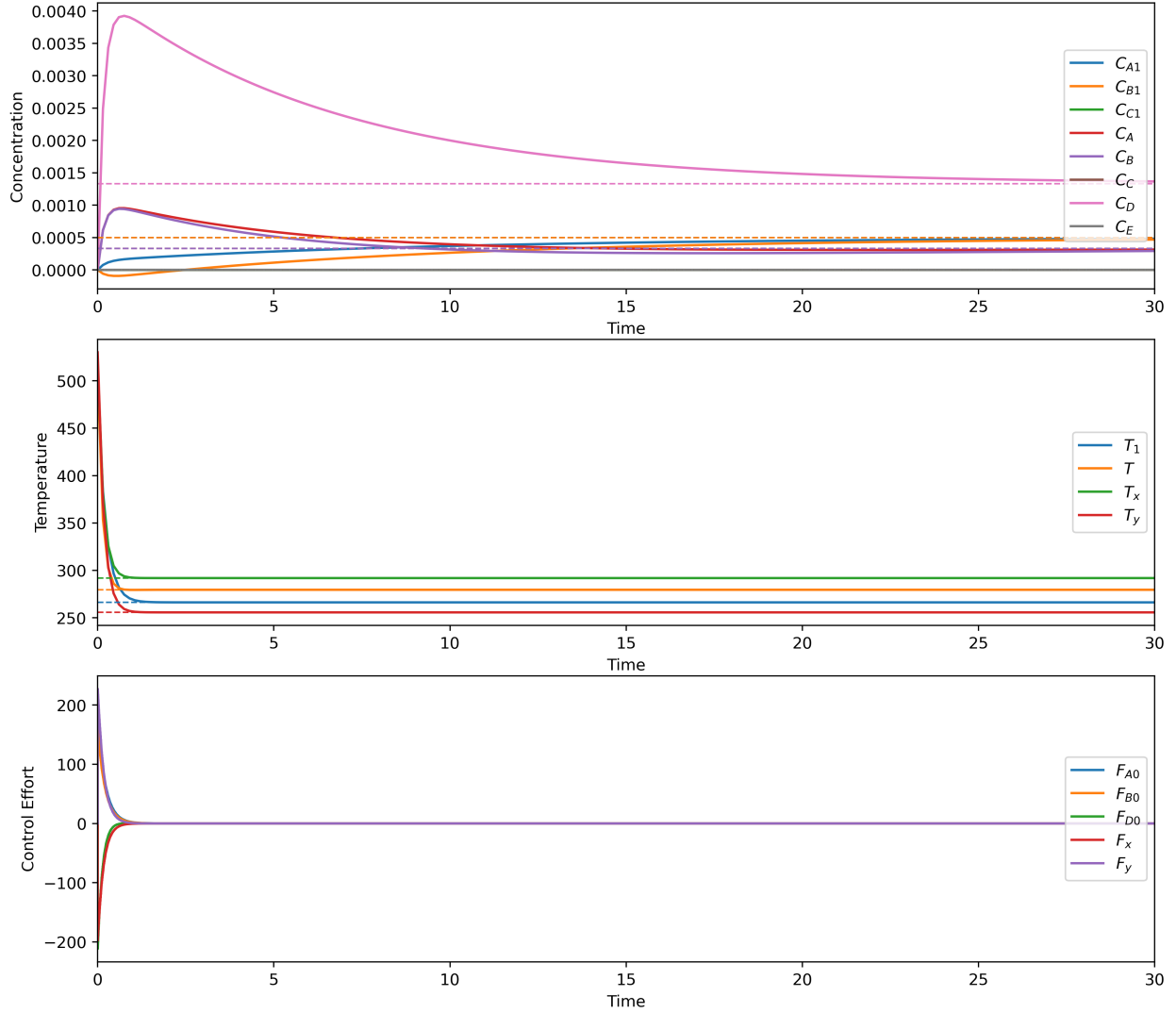


Figure 3: Concentration and temperature profiles for each species under LQR control.

other words, LQR does not care that it's essentially telling the flow rate to go backwards, which is why we see concentration going negative, for example.

One way that we can force the system to have more realistic dynamics is to set hard limits on the concentration/temperature/flow rate values; an illustrative example:

```
# Limit concentration from going over max or going under min
if CA1[i] > CA1_max:
    CA1[i] = CA1_max
elif CA1[i] < CA1_min:
    CA1[i] = CA1_min
```

This is essentially what is done when a controller becomes saturated, at which point the control system is now operating in an open-loop configuration. In the next section we will see what happens when we do saturate the controller by limiting the flow rate to be bounded between 0 and some F_{max} .

4 Experimenting with Controller Saturation

We will briefly investigate the event when the controller is saturated. This happens when applying an unconstrained control method to a system with actuator constraints, and the controller is attempting to apply a control effort that goes beyond the actuator's capabilities. When the actuator limit is reached, the controller is now attempting to control a system that is not responding to its inputs, and the loop is now broken. It isn't until the state reaches a point where the controller effort is within the actuator's capabilities that the loop closes again, and the controller is able to drive the state towards the origin. Applying this to the CSTR simulation yields the results in figure (4).

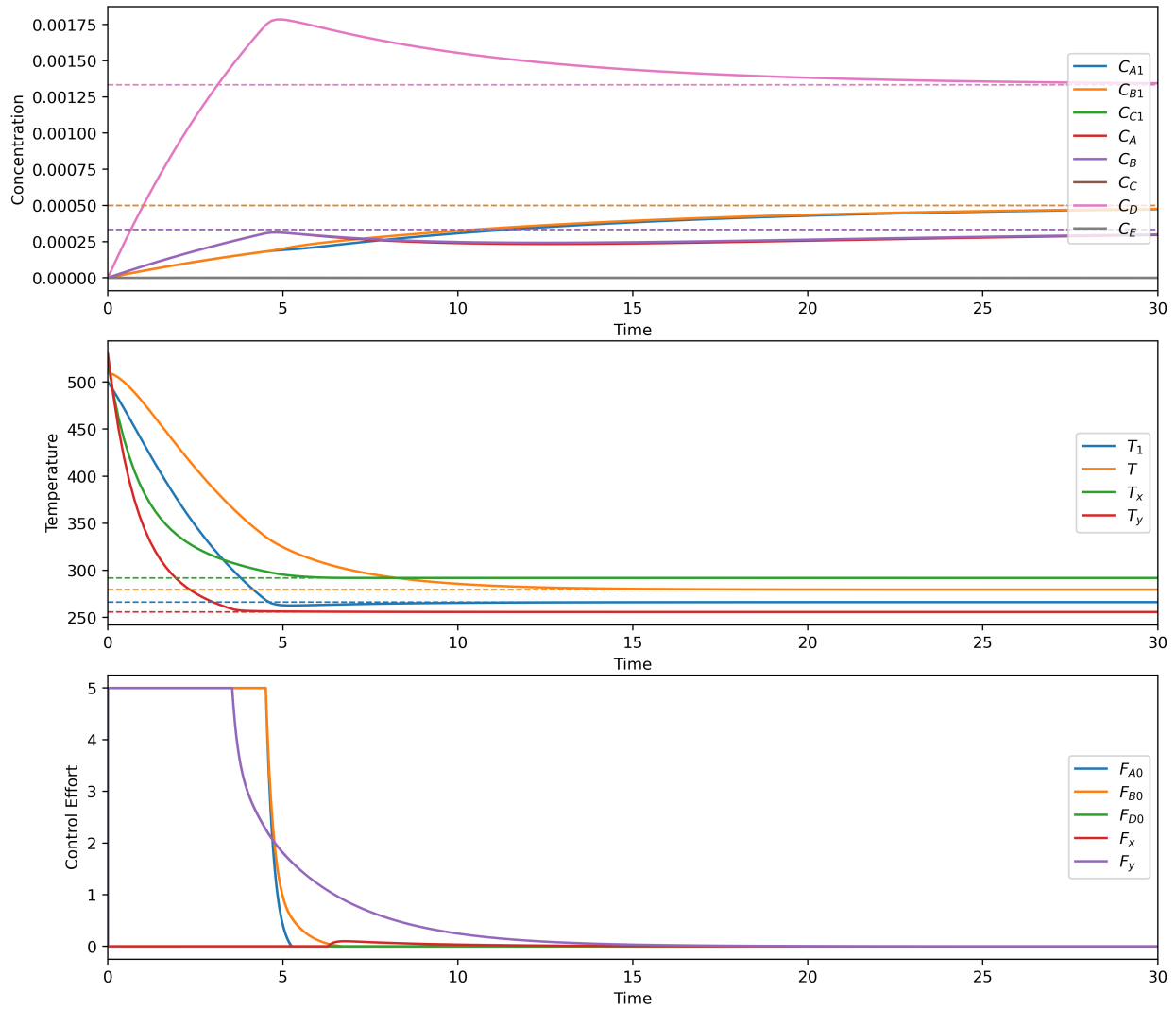


Figure 4: Concentration and temperature profiles for each species under saturated LQR control.

5 Remarks

There are just a few remarks that should be mentioned about determining values of parameters for simulation and programming the solution that could be useful to consider down the road.

- If parameters lie within “unrealistic” territory (perhaps the combination of activation energy and frequency factor make it so that the ODE solution results in a negative concentration), they *must* be reconciled for the model to make any meaningful sense. If possible, in the future, find real-world reactions and their measured parameters so that you are sure that the simulation is valid. I was banging my head against the wall for hours because the ODE solutions were yielding negative concentration, and I believe the solution to that was to modify the frequency factors and activation energies so that they weren’t “counteracted” by a high volumetric flow rate.
- Expanding on this a little bit, numerical integrators don’t “know” when the results are meaningless, and they will integrate like nothing happened. It is up to us to choose valid parameters to help bound the solution to within a meaningful range.
- `scipy.integrate.solve_ivp` is able to solve systems of ODEs in state-space form. This is extremely handy.

References

- [1] M. Shyamalagowri and R. Rajeswari, “Modeling and simulation of non linear process control reactor - continuous stirred tank reactor,” *International Journal of Advances in Engineering & Technology*, vol. 6, Sept 2013.

6 Appendix

6.1 Parameter Descriptions and Values

Table 1: Table of parameters, their descriptions, values (if applicable), and units that were used in simulating the dual CSTR system.

Parameter	Description	Value	Units
C_{A1}^{init}	Initial concentration of A in CSTR 1	0	mol/m ³
C_{B1}^{init}	Initial concentration of B in CSTR 1	0	mol/m ³
C_{C1}^{init}	Initial concentration of C in CSTR 1	0	mol/m ³
C_A^{init}	Initial concentration of A in CSTR 2	0	mol/m ³
C_B^{init}	Initial concentration of B in CSTR 2	0	mol/m ³
C_C^{init}	Initial concentration of C in CSTR 2	0	mol/m ³
C_D^{init}	Initial concentration of D in CSTR 2	0	mol/m ³
C_E^{init}	Initial concentration of E in CSTR 2	0	mol/m ³
C_{A0}	Inlet concentration of A	0.001	mol/m ³
C_{B0}	Inlet concentration of B	0.001	mol/m ³
C_{D0}	Inlet concentration of D	0.002	mol/m ³
T_{x0}	Inlet temperature of water x	298	Kelvin
T_{y0}	Inlet temperature of water y	250	Kelvin
F_{A0}	Initial inlet volumetric flow rate of A	0	m ³ /s
F_{B0}	Initial inlet volumetric flow rate of B	0	m ³ /s
F_{D0}	Initial inlet volumetric flow rate of D	0	m ³ /s
F_{x0}	Initial inlet volumetric flow rate of x	0	m ³ /s
F_{y0}	Initial inlet volumetric flow rate of y	0	m ³ /s
V_1	Volume of CSTR 1	20	m ³
V_2	Volume of CSTR 2	20	m ³
V_x	Volume of heat jacket around CSTR 1	1	m ³

V_y	Volume of heat jacket around CSTR 2	1	m^3
A_1	Frequency factor for R_1	1×10^7	$\text{m}^3\text{mol}^{-1}\text{s}^{-1}$
A_2	Frequency factor for R_2	1×10^8	$\text{m}^3\text{mol}^{-1}\text{s}^{-1}$
E_1	Activation energy for R_1	4×10^4	J/mol
E_2	Activation energy for R_2	1×10^4	J/mol
R_g	Ideal gas constant	8.314	$\text{J mol}^{-1}\text{K}^{-1}$
ΔH_1	Enthalpy of reaction for R_1	1×10^5	J/mol
ΔH_2	Enthalpy of reaction for R_2	5×10^5	J/mol
U_x	Overall heat transfer coefficient for heat jacket around CSTR 1	100	$\text{W m}^{-2}\text{K}^{-1}$
U_y	Overall heat transfer coefficient for heat jacket around CSTR 2	100	$\text{W m}^{-2}\text{K}^{-1}$
A_x	Heat transfer area for heat jacket around CSTR 1	10	m^2
A_y	Heat transfer area for heat jacket around CSTR 2	10	m^2
C_{pr}	Heat capacity of reactive mixture	4.184	J/K
C_{pc}	Heat capacity of heat jacket water	4.184	J/K
ρ_r	Density of reactive mixture	1000	kg/m^3
ρ_c	Density of heat jacket water	1000	kg/m^3

6.2 Steady-State Values

$$\mathbf{x}_{eq} = \begin{bmatrix} C_{A1}^{eq} \\ C_{B1}^{eq} \\ C_{C1}^{eq} \\ C_A^{eq} \\ C_B^{eq} \\ C_C^{eq} \\ C_D^{eq} \\ C_E^{eq} \\ T_1^{eq} \\ T^{eq} \\ T_x^{eq} \\ T_y^{eq} \end{bmatrix} = \begin{bmatrix} 0.000499646 \\ 0.000499646 \\ 3.54441 \times 10^{-7} \\ 0.00033285 \\ 0.00033285 \\ 4.02354 \times 10^{-11} \\ 0.00133285 \\ 4.83665 \times 10^{-7} \\ 266.249 \\ 279.527 \\ 291.875 \\ 255.696 \end{bmatrix}$$